

Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka

Probabilistic Assessment and Forecasting of Landslide Occurrences

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka has two monsoon seasons every year. These seasons often bring rains that cause landslides in the central highlands. The landslides lead to loss of life people being displaced from their homes and damage to roads, bridges and buildings. One area that sees these problems repeatedly is the Nuwara Eliya District. Within this district the Walapane, Ambagamuwa and Kotmale areas are especially vulnerable. Despite this these regions have not been studied much as other high-risk zones like Sabaragamuwa Province. Previous studies show that the accuracy of landslide prediction models depends heavily on the data used. Different types of landslide records levels of detail in elevation maps and the number of factors considered can change model performance by as much as 0.2 in AUC scores. Also simple methods such as regression sometimes work better than complex machine-learning models when predicting landslides outside the training area. In Sri Lanka combining rainfall, slope, elevation, soil type, land cover and distance to rivers using the Analytic Hierarchy Process improved prediction accuracy. Models based on rainfall reached **50% to 70%** accuracy.. Adding multiple factors pushed accuracy up to 86%. Still these improvements relied mostly on judgment for assigning weights to each factor. They did not use machine learning to compare options

There is a gap here. No system currently exists that gives time accurate probability forecasts for landslides in Nuwara Eliya. There is also no early-warning tool tailored to local needs. To address this this paper introduces a component called the Probabilistic Landslide Occurrence Assessment and Forecasting module. This module is part of a system called the **Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka**. The new module uses data like **rainfall, soil type, historical landslide locations and slope angles derived from elevation models (DEMs)**. It calculates the chance of a landslide occurring. Classifies risk into five levels: Very Low, Low, Moderate, High and Very High. It compares machine-learning models. Including XGBoost and LightGBM. Against simpler models. These models are tested using -fold cross-validation. Historical landslide data from the National Building Research Organisation (NBRO) is used to validate results.

The design also includes a Web-GIS interface that supports both Sinhala and Tamil. This makes it easier for local users to access information. The interface uses high-contrast visuals to help older adults and people in rural areas. It also includes an explainable-AI layer using SHAP and LIME techniques. This helps users understand what factors contributed to each prediction. Such as slopes, recent rain or certain soil types.

the outputs from this module. Both the probability and severity estimates. Feed directly, into another component. That component focuses on impact and decision support. This allows emergency teams to act quickly and allocate resources where they are most needed. Ultimately the goal is to reduce risks and save lives through data-driven response systems.

Keywords - landslide susceptibility, machine learning, probabilistic forecasting, explainable AI, disaster risk reduction, Enhanced Technology

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1. INTRODUCTION

1.1 Background and Context

The real world problem. Sri Lanka's central highlands experience two monsoon seasons each year the southwest monsoon (May–September) and the northeast monsoon (October–January) both of which saturate steep, weathered terrain and regularly trigger landslides, cutting failures, and slope failures. These events destroy homes, sever roads and utilities, and in the worst cases cause mass casualties, with damage that compounds each year as climate variability intensifies rainfall extremes [4].

Who is affected, and why it matters. The most exposed communities are not in Sri Lanka's major cities but in rural, plantation-dependent hill-country divisions in Nuwara Eliya District, specifically the Walapane, Ambagamuwa, and Kotmale Divisional Secretariat (DS) divisions. These areas combine steep slopes, high annual rainfall, and dense population settled directly on or below hazard-prone terrain, including estate labour communities with limited resources to relocate or rebuild after a disaster. The National Building Research Organisation (NBRO) issues district- and divisional-level landslide early warnings for these exact divisions during nearly every monsoon [5], underscoring how routine and how under-resourced the exposure is.

Statistical evidence.

Global: Landslides caused over 18,000 deaths and affected 4.8 million people worldwide between 1998 and 2017, with Asia accounting for 54% of all recorded events and global economic losses reaching approximately USD 8 billion [3].

Local: Sri Lanka's central highland region recorded nearly 1,000 landslide-related deaths between 2003 and 2016 [3]. In Kegalle District alone, 13% of the land area was found to be highly susceptible to landslides and 37% prone to slope failure as of 2018 [3]. For Nuwara Eliya specifically, NBRO's own risk-profiling work has flagged Walapane and neighbouring divisions as recurring high-exposure zones, with warnings issued for Kotmale, Walapane, Hanguranketha, and Ambagamuwa in the same advisory nearly every rainy season [5].

Analysis of existing solutions and their limitations. Current landslide-risk approaches fall into three broad categories, each with a specific shortfall relevant to this component:

1. Heuristic, expert-weighted zonation (AHP-based). Studies conducted directly in Nuwara Eliya's Walapane division have produced landslide hazard zonation maps using the Analytic Hierarchy Process across six static factors slope, rainfall, soil type, land use, geology, and land-use change classified into fixed low/moderate/high risk bands [5]. A similar AHP approach applied in Kegalle District improved accuracy from a 50–70% rainfall-only baseline to 86% by incorporating five additional factors [3]. These approaches are valuable and locally grounded,

2. Data-driven statistical/ML susceptibility mapping (international literature). Research using large landslide inventories has shown that model accuracy depends heavily on the type of inventory used (manual vs. automatic), the resolution of the topographic data, and the number of causative factors included with AUC varying by as much as 0.2 across similar models [1]. Separate work has shown that simpler classifiers such as logistic regression often generalize better outside their training area than more complex machine-learning methods like SVM or bootstrap-aggregated trees, which tend to overfit [2]. This body of work establishes a sound comparative methodology, but it has been developed and validated in Mexico and Ecuador respectively, not calibrated to Sri Lankan terrain, soils, or monsoon rainfall behaviour, and none of it is built for real-time forecasting.

Taken together, no existing solution heuristic, data-driven, or operational combines locally-calibrated, comparative machine learning with a real-time occurrence probability, transparent factor-level explanation, and an interface genuinely accessible to the communities who receive these warnings. This is the gap the proposed component addresses.

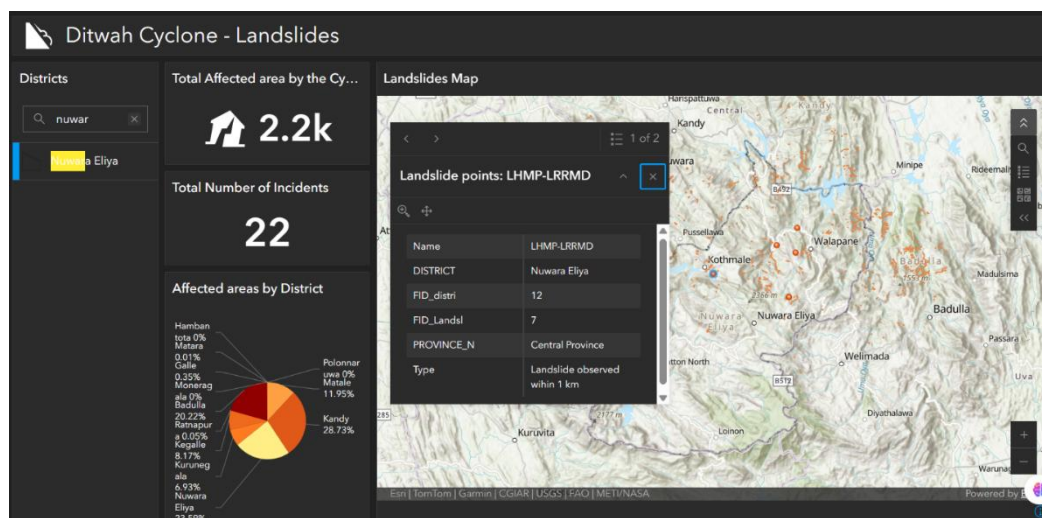


Figure 1. Landslide incident distribution in Nuwara Eliya District following Cyclone Ditwah, showing 22 recorded incidents concentrated around Walapane and Kotmale — the study area's most exposed divisions. Source: [dashboard publisher]

Proposed solution and its parts. This component addresses that gap with three integrated parts: (1) a multi-factor probability engine that correlates rainfall, soil type, historical landslide inventory, and slope angle (derived from DEM data), comparatively evaluating multiple ML models to generate a landslide occurrence probability value for each relevant area, which is then classified into five susceptibility tiers Very Low, Low, Moderate, High, and Very High for interpretability; (2) an inclusive, dual-language (Sinhala/Tamil) Web-GIS live mapping interface with high-contrast, accessible themes for elderly and village-level users; and (3) an explainable-AI layer (SHAP/LIME) that breaks down each prediction into the specific environmental and human factors driving it, moving from broad regional alerts toward site-specific, actionable warnings.[8]

SDG alignment. This work aligns primarily with **SDG 11 (Sustainable Cities and Communities)**, specifically Target 11.5, which calls for significantly reducing deaths and economic losses from disasters, and **SDG 13 (Climate Action)**, Target 13.1, which calls for strengthening resilience and adaptive capacity to climate-related hazards. It also contributes indirectly to **SDG 3 (Good Health and Well-being)** by reducing disaster-related casualties, and **SDG 9 (Industry, Innovation and Infrastructure)** through resilient, technology-enabled infrastructure planning.

Research cluster alignment. This component aligns with SLIIT's **Center of Excellence for AI (CEAI)**, a multidisciplinary initiative advancing applied AI research across the university. The component's core contribution a comparative evaluation of multiple machine-learning models for probabilistic disaster forecasting, paired with an explainable-AI layer (SHAP/LIME) that renders model predictions transparent and interpretable sits squarely within CEAI's focus on applied, real-world AI systems rather than purely theoretical model development. By grounding the research in a locally-calibrated, high-stakes application domain (landslide occurrence prediction for a specific, historically underserved district), this component also reflects CEAI's broader emphasis on AI research that produces measurable societal impact, consistent with the multidisciplinary, solutions-oriented approach the center is built around.

1.2 Literature Review

Part A: Peer-Reviewed Literature

A.1 Gaidzik & Ramírez-Herrera (2021) - "The Importance of Input Data on Landslide Susceptibility Mapping"

- **Method:** Logistic regression was used to build 32 separate landslide susceptibility models, systematically varying four input factors: landslide inventory type (manual vs. automatic), DEM resolution (15 m vs. 1 m), number of causative factors (12 vs. 15), and landslide sampling technique (mass vs. mass-center) - to isolate which most strongly affects model accuracy.
- **Dataset:** Two study regions in the Coyuca River basin, southwest Mexico (15.6 km² and 6.7 km²); landslide inventories built from both manual satellite-image interpretation (419 landslides) and an automated Contour Connection Method (1,726 features), calibrated against 15 m and 1 m resolution DEMs.
- **Context:** Tropical, hurricane-prone, mountainous terrain geomorphologically comparable to Sri Lanka's central highlands, though on a different continent and under a different rainfall regime.
- **Findings:** Models based on automatic landslide inventory achieved comparable overall prediction accuracy to those built from manually detected features, and finer-resolution topographic data produced more accurate and precise susceptibility models. Sampling from the full landslide mass rather than just its center point consistently produced more reliable models, with AUC values differing by up to 0.2 depending on configuration.
- **Strengths:** A rigorous, controlled comparison that isolates single variables genuinely useful for justifying *why* specific model design choices (inventory type, resolution, sampling method) matter, and it provides defensible accuracy benchmarks to compare this component's own results against.
- **Limitations relative to this research:** A single-algorithm study (logistic regression only) with no comparative ML evaluation; not designed for real-time use; not calibrated to a Sri Lankan or tropical-monsoon-village context; produces static maps rather than a live, continuously updated probability feed.
- **Influence on proposed research:** This study directly motivates the comparative multi-model design (XGBoost, LightGBM, logistic regression) and the emphasis on data-quality decisions: inventory source, resolution, sampling as first-class methodological choices rather than afterthoughts.

A.2 Brenning (2005) - "Spatial Prediction Models for Landslide Hazards: Review, Comparison and Evaluation"

- **Method:** Comparative evaluation of logistic regression (with and without stepwise variable selection, and with spatial-autocorrelation modelling), support vector machines, and bootstrap-aggregated classification trees (bagging and double-bagging), using a spatial cross-validation scheme that separates test and training points by an empirically derived distance buffer to avoid spatial-autocorrelation bias.
- **Dataset:** The Estación Científica San Francisco area in the Andes of southern Ecuador (11.2 km²); a multi-temporal landslide inventory compiled across five survey years (1962–1998), enabling both "present" and "future" prediction scenarios.
- **Context:** Tropical mountain rainforest terrain with slashing/burning and grazing-driven land-use change a socio-environmental profile with parallels to human-driven landslide triggers (road cuts, deforestation) relevant to Nuwara Eliya.
- **Findings:** Logistic regression with stepwise variable selection achieved the lowest error rates and the best generalization outside the training area of all methods tested, while tree-based methods (bagging, double-bagging) substantially overfit the training data despite near-zero apparent error rates on the training set itself.
- **Strengths:** Provides strong, citable evidence that logistic regression is not a naive or outdated baseline but a genuinely competitive method with superior generalization properties - directly justifying its inclusion in this component's comparative model set rather than treating it as a formality.
- **Limitations relative to this research:** Two-decade-old dataset and toolset, predating gradient-boosted methods such as XGBoost and LightGBM entirely; no explainability layer of any kind; a research case study with no path to operational or real-time deployment.
- **Influence on proposed research:** Justifies retaining logistic regression as a required comparison baseline (not just a placeholder) and reinforces the importance of rigorous, spatially-aware validation (this component will need to guard against similar overfitting risks, particularly if XGBoost/LightGBM are tuned aggressively on a comparatively small Nuwara Eliya inventory).

A.3 Gunasinghe, Gunawardhana & Rajapakse (2023) - "Predictive Analysis of Landslide Susceptibility under Hydrological Aspects of Climate Change in Kegalle District, Sri Lanka"

- **Method:** Analytic Hierarchy Process (AHP), combining six expert-weighted factors rainfall-threshold exceedance days, slope, elevation, distance from river, soil type, and land use/land cover into a single weighted susceptibility score; future rainfall projected via the HadGEM3-GC31-LL climate model (CMIP6), statistically downscaled using LARS-WG.
- **Dataset:** A 1,523 km² catchment in Kegalle District, Sri Lanka; historical landslide, slope-failure, and cutting-failure records from NBRO; CHIRPS gridded rainfall data (5 km × 5 km resolution).
- **Context:** Sri Lanka's central highland region, directly adjacent to and climatically similar to Nuwara Eliya District, making it the most geographically relevant of the three literature sources.
- **Findings:** Rainfall alone matched 50–70% of observed landslide occurrences; incorporating the additional five factors raised overall model accuracy to 86%, with 25 of 29 validation events falling into "high" or "very high" risk zones. Future climate scenarios (SSP2-4.5, SSP5-8.5) project substantial expansion of high-risk zones, with high/very-high susceptibility area coverage rising from 46.95% (observed baseline) to as much as 72.97% under SSP5.
- **Strengths:** Directly local to Sri Lanka; demonstrates that multi-factor modelling substantially outperforms rainfall-only approaches in exactly the terrain and climate this component targets; provides a credible accuracy benchmark (86%) against which this component's ML models can be compared.
- **Limitations relative to this research:** Heuristic, expert-assigned AHP weights rather than weights learned from training data; no comparative evaluation across multiple algorithms; produces static seasonal maps rather than a continuously updated probability; not validated specifically for Nuwara Eliya's Walapane/Ambagamuwa/Kotmale terrain.
- **Influence on proposed research:** Confirms that multi-factor approaches are essential in the Sri Lankan context and gives a concrete accuracy target (86%) to benchmark against; motivates replacing subjective AHP weighting with models that learn factor importance directly from historical Nuwara Eliya landslide data.

Part B :Existing Systems and Products

B.1 ResQConnect (Aththanayake et al., 2026)

- **Functions:** An agentic Retrieval-Augmented Generation (RAG) pipeline that converts citizen help-request text into structured, SOP-grounded task plans; an adaptive event-triggered multi-commodity vehicle-routing engine for relief-resource distribution; a compressed, edge-deployed language model providing offline safety guidance on smartphones when connectivity fails.
- **Technology:** A multi-agent LLM architecture (dedicated metadata-extraction, retrieval, adequacy-assessor, and task-planner agents) built on GPT-4o for the cloud pipeline; a quantized 0.5-billion-parameter model (Qwen2.5-0.5B) fine-tuned on a curated disaster Q&A dataset and converted to MediaPipe's .task format for on-device inference.
- **Performance:** Improved overall task-quality scores from 61.4 to 82.9 (+21.5 points) over a standard RAG baseline; reduced routing solver calls by up to 85% compared to continuous re-optimization while remaining within 7–12% of optimal response time; delivered fully offline mobile guidance with sub-500 ms response latency and 54 tokens/s throughput on a commodity smartphone (Samsung Galaxy S23 Ultra).
- **Users:** Disaster-management agencies and field responders (task breakdowns, routing plans) and affected citizens (mobile chatbot for offline guidance).
- **Limitations relative to this component:** ResQConnect operates entirely on the response side of the disaster timeline - it assumes a hazard has already occurred and been reported by a citizen, then optimizes coordination, logistics, and guidance. It contains **no predictive or probabilistic hazard-occurrence capability at all**; floods and landslides are categories within its knowledge base, not phenomena it forecasts. It is also validated on aggregated national-scale Sri Lankan disaster data rather than calibrated to a specific district's terrain, soil, and historical landslide inventory.

B.2 NBRO/DMC Operational Early-Warning System

- **Functions:** District- and DS-division-level landslide and flood advisories, rainfall/water-level bulletins, and situation reports, issued multiple times daily during the monsoon season.
- **Technology:** Threshold-based triggers (e.g., the established 73 mm/day rainfall threshold used in NBRO's own landslide-triggering criteria) rather than a trained predictive model; manually compiled and distributed bulletins.
- **Performance:** No published model-accuracy metric (it is an advisory service, not a predictive model); operational responsiveness is high, with bulletins reissued several times per day during active weather events.
- **Users:** The general public, district and divisional disaster-management officials, and downstream agencies such as the DMC.

- Limitations relative to this component: Fundamentally reactive and threshold-based rather than predictive; granularity stops at the district or DS-division level, well short of a specific slope or settlement; provides no probability value, no severity tier, and no explanation of which factors are driving the risk.

B.3 Walapane AHP Landslide Hazard Zonation Study (2018)

- Functions: A static, seasonal landslide hazard zonation map for Walapane DS division the same DS division targeted by this component classifying land into low/moderate/high risk.
- Technology: GIS-based Analytic Hierarchy Process combining six factors: slope, annual rainfall variation, soil type, land-use change, geology, and observed landslide timing patterns.
- Performance: Risk classification produced for three high-risk months (November, December, January) based on historical landslide timing; no AUC or percentage-accuracy figure reported.
- Users: Land-use planners and local disaster-management authorities in Walapane.
- Limitations relative to this component: No machine-learning component and no comparative model evaluation; static, seasonal output rather than a continuously updated probability; despite being the most geographically precise prior study (Walapane itself), it predates the more comprehensive multi-factor approaches later validated in Kegalle [3] and offers no explainability or accessible public-facing interface.

1.3 Research Gap

Across both the peer-reviewed literature and the operational systems reviewed in Section 1.2, a consistent pattern emerges: every existing approach is strong in exactly one of three dimensions predictive rigor, real-time accessibility, or interpretability but no single system, published study, or operational tool combines all three for a locally calibrated landslide-occurrence model in Sri Lanka's central highlands. This proposed component is structured around three integrated parts specifically because each one closes a distinct, evidenced gap left open by the systems already discussed.

Feature 1- No comparative, locally calibrated probabilistic forecasting engine. The strongest predictive methodologies in the literature the systematic input-data comparison by Gaidzik and Ramírez-Herrera [1] and the multi-classifier evaluation by Brenning [2] are rigorous but were developed and validated in Mexico and Ecuador respectively, using terrain, soil, and rainfall regimes unrelated to Sri Lanka, and neither compares more than a handful of classical statistical methods against modern gradient-boosted approaches. Conversely, the Sri Lankan studies that are locally calibrated the Kegalle AHP model [3] and the Walapane zonation study [5] rely on subjectively assigned expert weights rather than models trained and validated against actual outcome data, and neither performs any comparative model evaluation at all. It is currently unknown which machine-learning approach (XGBoost, LightGBM, or logistic regression) produces the most accurate landslide-occurrence probability when trained specifically on Nuwara Eliya's terrain, soil, rainfall, and historical inventory no publication or system reviewed answers this question for this district.

Feature 2- No continuously updated, accessible severity output at the community level. Operationally, NBRO and DMC bulletins are issued at the district or DS-division level [4], and the AHP-based studies [3, 5] produce static, seasonal susceptibility maps rather than a live, continuously updated value. Neither delivers a specific probability or severity tier to a specific village, nor presents it through an interface designed for the people receiving the warning elderly and rural residents in Sinhala- or Tamil-speaking communities. ResQConnect [4] comes closest to a citizen-facing interface, but only after an incident has already been reported; it has no mechanism for surfacing a hazard probability *before* an event occurs. **It is inadequately addressed how a real-time landslide probability and severity tier can be surfaced directly and accessible to the at-risk communities themselves**, rather than only to district-level officials.

Feature 3- No factor-level explanation of individual predictions. Every susceptibility model reviewed heuristic (AHP-based [3, 5]) or statistical (logistic regression, SVM, tree-based [1, 2]) outputs either a fixed weight or a single aggregate probability, with no mechanism for showing *why* a specific location is at risk. None decomposes a prediction into the relative contribution of environmental factors (rainfall, soil, slope) versus human factors (road cuts, deforestation, drainage blockage) for that specific site. **This is currently unknown at the individual-prediction level in every system reviewed**, and it is precisely the shift from broad regional risk to explainable, site-specific causation that separates a warning from actionable knowledge.

2. OBJECTIVE

2.1 Main Objective

To create an Intelligent and Integrated disaster management system that identifies flood and landslide risks, estimates their impacts, and provides timely decision support and emergency response recommendations for Sri Lanka.

2.2 Specific Objectives

To acquire and compile the core landslide dataset for Nuwara Eliya District historical landslide point locations, NBRO hazard zones, and high-resolution precipitation records through formal data requests to national departments.

To construct a multi-factor geospatial feature set by layering DEM-derived slope angle, bedrock/soil classification, and rainfall trigger thresholds (based on official NBRO alert levels) within a spatial database backend.

To integrate human-factor risk parameters including slope cutting, deforestation, and unplanned drainage obstruction as supplementary predictive inputs alongside the environmental factors.

To train and comparatively evaluate machine-learning classification models (XGBoost, LightGBM, and logistic regression as baseline) on the combined feature set, achieving granular spatial triage from provincial scale down to individual towns and villages.

To develop a WebGL-powered, dual-language (Sinhala/Tamil/English) live mapping interface with accessible, high-contrast themes, rendering dynamic terrain-stability indicators for elderly and village-level users.

To implement an explainable-AI layer (SHAP/LIME) that decomposes each site-specific prediction into its percentage contribution from environmental versus human causative factors, closing the gap between broad regional warnings and site-specific evaluation.

3. METHODOLOGY

3.1 Research Methodology / Design

Nature of the research question: The core question “can a machine-learning model predict landslide occurrence probability more accurately than existing heuristic methods for Nuwara Eliya District?” is a comparative, quantifiable question with a measurable ground truth (documented landslide occurrence vs. non-occurrence). It is not exploratory or interpretive; it requires numeric, testable answers.

Type of data: All inputs are structured, numeric or categorical geospatial data rainfall records, DEM-derived slope values, categorical soil classes, and binary/multi-class historical incident labels. This is precisely the data shape quantitative predictive modelling is designed for and is structurally identical to the datasets used in [1], [2], and [3][10].

Experimental requirements: Comparing multiple classifiers (XGBoost, LightGBM, logistic regression) against each other and against existing benchmarks requires a controlled experimental setup fixed train/test partition, consistent evaluation metrics across models, and repeatable runs. This rules out purely qualitative or case-study designs.

Relevant literature: Both Gaidzik and Ramírez-Herrera [1] and Brenning [2] use this same design multiple models or configurations trained and evaluated under identical conditions, with performance compared using standard classification metrics (AUC, error rate). This proposal follows the same established convention rather than departing from it.

Selected methodology: A **quantitative, experimental (comparative) research design**, structured as a supervised machine-learning modelling pipeline: data acquisition → feature engineering → comparative model training → statistical evaluation → validation against external benchmarks.

3.2 Data / Experimental Inputs

This component involves no human participants it is a geospatial/statistical modelling task using secondary administrative and environmental data. (Note: if the mapping interface in Objective 5 undergoes later usability testing with village-level users, that would introduce a small participant-based sub-study; this is addressed separately under Section 3.4 if applicable.)

- **Data sources:** The official NBRI geospatial data package (NBRI-GEO-PKG-2026) obtained through a formal data-supply agreement with the National Building Research Institute, supplemented by NBRO rainfall and precipitation records, DEM-derived slope data, and a soil classification layer sourced from the team's shared dataset repository [8].
- **Dataset composition:**

- 1:50,000 Landslide Hazard Zonation Map (LHMP) 3,083 polygons, classified into hazard Range classes 1–4.
- Historical incident inventory by failure type across the target DS divisions: Cutting Failure (940), Landslide (81), Rock Fall (83), Slope Failure (56), Ground Subsidence (30), Other (15).
- Post-disaster high-risk inspection points from Cyclone Dithawa (507 points, classified HR1/HR2/HR3).
- Mapped landslide footprint polygons (268 features).

Rainfall threshold/intensity records and DEM-derived slope angle for the same divisions.

- **Study area / sample:** Walapane, Ambagamuwa, and Kotmale Divisional Secretariat divisions within Nuwara Eliya District, following the same DS-division-level scoping used in the Kegalle study [3] and the Walapane-specific AHP study [5].
- **Collection method:** Formal institutional data-sharing agreements with NBRO/NBRI and the Disaster Management Centre (already in progress); supplementary layers retrieved from the team's shared GitHub dataset repository [8].
- **Preparation process:** Following the sampling convention validated in [1], positive (landslide) and negative (non-landslide) points will be extracted in equal number to avoid class-imbalance bias. Geospatial preprocessing (QGIS/PostGIS) will align all layers to a common coordinate reference system, extract slope from the DEM, encode soil type categorically, compute rainfall-threshold-exceedance days per the NBRO 73 mm/day convention established in [3], and spatially join incident records to their corresponding feature values before the data is split into training and test partitions.

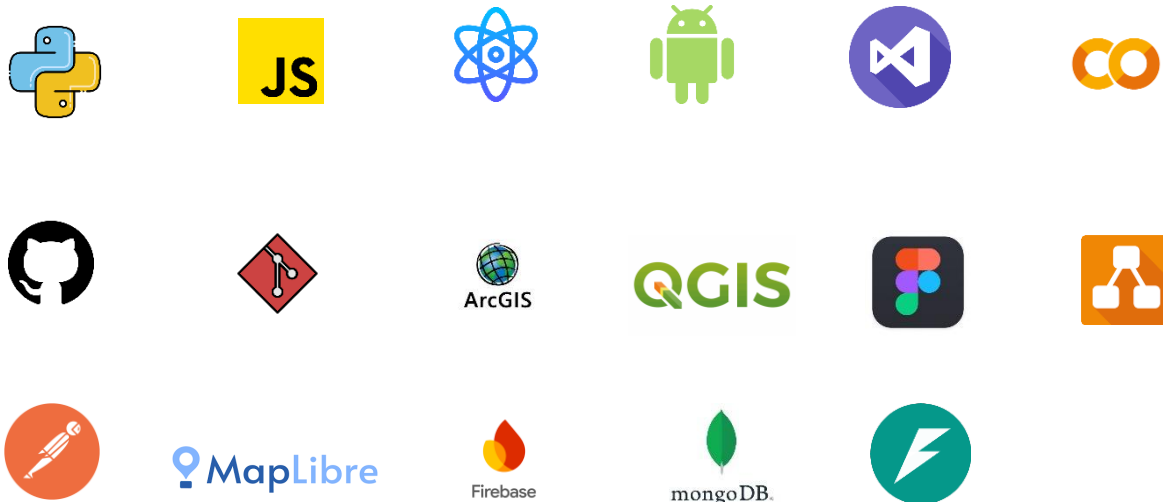


Figure 2. Tools and Software That We Going to Use to Implement Our Research Components

3.3 Baseline / Benchmark

1. **Heuristic AHP benchmark** - the Kegalle multi-factor AHP model achieved 86% accuracy [3]; the Walapane-specific AHP zonation study [5] represents the most geographically precise (but non-ML) prior benchmark.
2. **Classical statistical baseline** - logistic regression, shown by Brenning [2] to generalize better outside its training area than more complex classifiers, and included in this study specifically to test whether that finding holds for Nuwara Eliya's terrain.
3. **International ML/DEM-resolution benchmark** - the AUC range (0.6–0.97 depending on configuration) reported by Gaidzik and Ramírez-Herrera [1] provides an external point of comparison for what "good" model performance looks like under varying data-quality conditions.

3.4 Evaluation Metrics

Comparison	Why Relevant	Target/Comparison
AUC-ROC	Threshold-independent measure of discrimination; the standard metric used across all three reviewed papers [1][2][3], enabling direct comparability	≥ 0.86 (matching/exceeding Kegalle AHP benchmark [3])
Accuracy (%)	Directly comparable to the Sri Lankan literature convention of reporting overall percentage accuracy	$\geq 86\%$
Precision / Recall / F1	Landslide datasets are inherently imbalanced (positive events are rare); accuracy alone can mask a model that misses most actual landslides. Recall is prioritized because a missed landslide (false negative) is more costly than a false alarm in a public-warning context [11]	Recall ≥ 0.80
Explainability consistency	Validates that the explainable-AI layer's factor attributions (Objective 6) are stable and reproducible rather than noisy artefacts of a single run	Consistent top 3 contributing factors across ≥ 5 repeated training runs
Inference latency	The component is intended to deliver a near-real-time probability, not a static seasonal map	< 1 second per query area on standard hardware

3.5 Validation Strategy

Following the spatial cross-validation approach used by Brenning [2] who showed that naive random train/test splits over-inflate accuracy on spatial data due to autocorrelation between nearby points this component will use k-fold cross-validation with spatial buffering, ensuring test points are not spatially adjacent to training points from the same landslide event.

Statistical significance between competing models will be assessed using a paired Wilcoxon signed-rank test across cross-validation folds, consistent with the comparative methodology already piloted in this project's earlier synthetic-data feature-set comparison. Where multiple comparisons are made (e.g., across all three models), a Bonferroni correction will be applied to control the family-wise error rate, following the precedent set in [2].

Finally, the model will undergo external validation against a held-out set of real, documented incidents specifically, post-Cyclone-Dithawa high-risk inspection points not used during training to test whether the model's predicted probabilities align with an actual, independently verified disaster event rather than only cross-validated historical data [12]

3.6 Development Approach

Selected approach: An Agile, iterative prototyping approach, structured around short sprints aligned with the CRISP-DM data-science lifecycle (understanding the geospatial → data understanding → data preparation → modelling → evaluation → deployment), repeated iteratively as new data arrives.

Justification: This is appropriate for three reasons specific to this project:

1. Staged data delivery - NBRO/NBRI/DMC data has arrived incrementally through formal agreements, meaning the feature set could not be finalized upfront; the project has already gone through one documented revision (from a slope-angle/DEM-centric feature set to a three-factor historical/rainfall/soil-centric one, then back to a four-factor comparison after empirical testing) exactly the kind of adaptive redesign Agile is built to accommodate rather than resist.
2. Empirical, test-driven model selection - comparative ML work is inherently iterative: each model configuration is a testable increment, evaluated and refined before the next is attempted, rather than designed fully in advance.
3. Integration with a four-person team building interdependent components. My component's probability and severity output feeds directly into a teammate's Decision Support component, requiring short, demonstrable increments (data pipeline → baseline model → comparative tuning → explainability layer → mapping interface) that can be integrated and tested against the evolving system rather than delivered as one large, late-stage unit.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The proposed system, Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka, is composed of four interdependent components that together transform raw environmental and citizen-generated data into coordinated disaster response actions for Sri Lanka. These are: (i) Flood Impact Assessment and Dynamic Mapping, which estimates flood severity and extent from hydrological and geospatial data; (ii) Landslide Prediction (this author's component), which computes site-specific landslide probability and identifies its causal factors; (iii) Multi-Hazard Early Warning Decision Support, which fuses hazard outputs from both prediction components into prioritized, actionable alerts; and (iv) Emergency Response and Resource Coordination, which converts alerts into dispatch and evacuation plans. The design follows a pipeline pattern comparable to recent human-centered disaster-response architectures, where specialized modules each handle one stage of the response chain and communicate through a shared, structured interface rather than a single monolithic model [ResQConnect].

4.2 System Diagram of Components



Figure 3. architecture diagram, black-and-white, own component highlighted

4.3 Individual Component

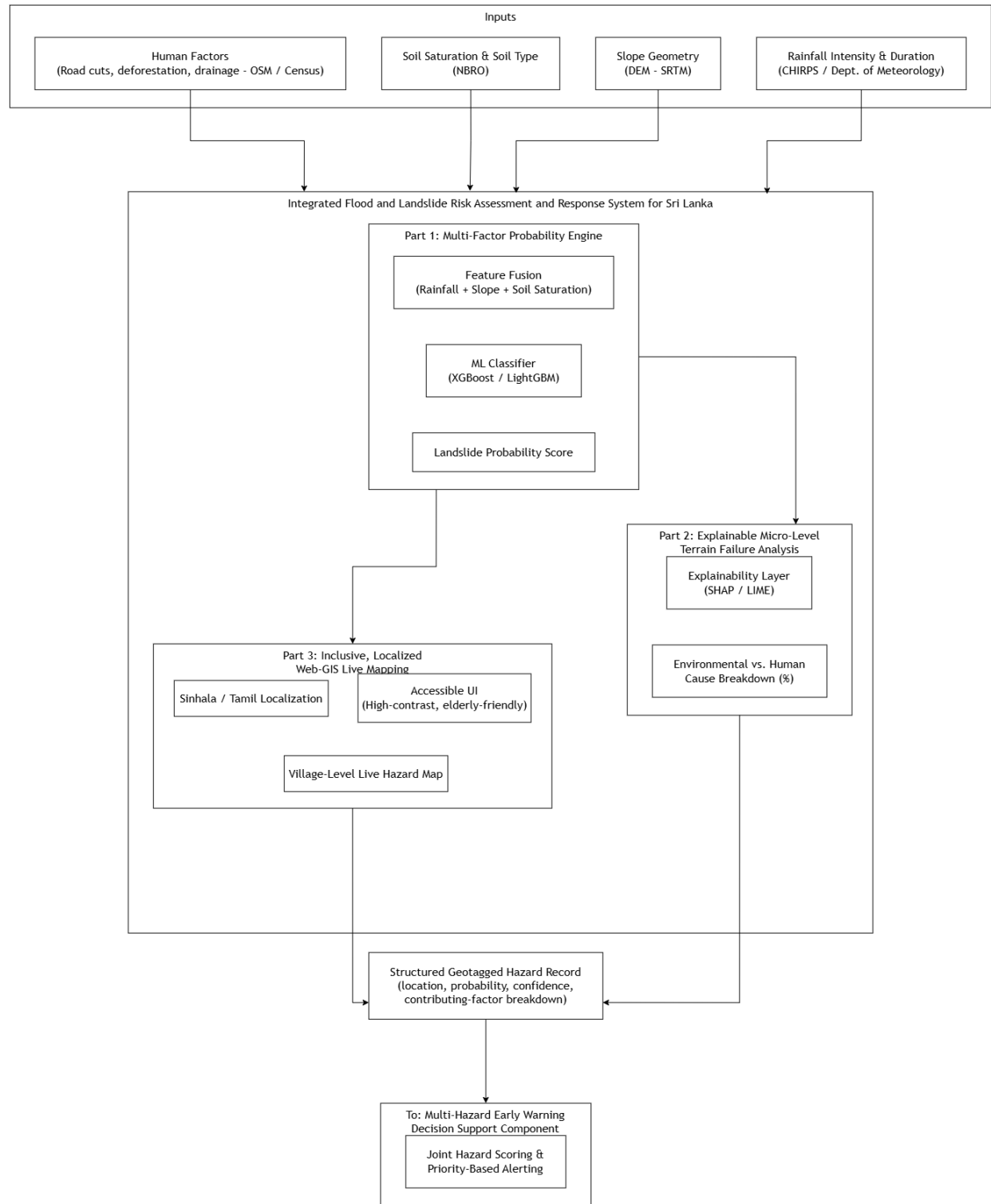


Figure 4. System Architecture Diagram Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka

4.4 Overall Integration

How the four components form a complete solution: Each component solves one link in the response chain, and none is independently sufficient the Flood and Landslide components only produce hazard likelihoods, the Decision Support component only prioritizes, and Emergency Response only acts. Only in sequence do they constitute an end-to-end early-warning-to-response system.

Dependencies: The Decision Support component has a hard input dependency on both the Landslide and Flood components' hazard outputs - it cannot compute a joint, multi-hazard risk score without both. Emergency Response, in turn, depends entirely on Decision Support's prioritized alert output; it does not consume raw hazard scores directly. The Landslide and Flood components are themselves independent of one another (no cross-dependency), which allows them to be developed, tested, and iterated in parallel.

Integration method: Components communicate through a standardized REST API contract, exchanging structured JSON records (following the geotagged hazard-record schema above) rather than sharing internal state, complemented by a common PostGIS spatial database for shared geospatial layers (DEM, administrative boundaries, road networks). This mirrors the metadata-driven, agent-boundary design used in ResQConnect, where each specialized module (metadata extraction, retrieval, routing) exchanges only a defined, minimal data contract with the next stage rather than exposing internal reasoning, which keeps components loosely coupled and independently upgradable [ResQConnect]. This integration approach was deliberately chosen so that this author's Landslide Prediction component can be developed and validated in isolation using historical/simulated data, then plugged into the shared pipeline once its output schema is finalized directly addressing the screening panel's feedback that overlapping responsibilities between components (particularly around Component 4) needed clearer separation, which this API-contract-based integration enforces structurally[13][14].

4.5 Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Multi-factor landslide probability model (rainfall + slope + soil saturation)	✓		
Explainable AI layer (SHAP/LIME) for terrain failure cause breakdown	✓		
Localized, accessible Web-GIS live mapping for landslide risk	✓		
Digitizing human factors (road cuts, deforestation) for pilot area	✓		
Data request to DMC/NBRO (landslide, decision support, emergency response components)		✓	
Shared PostGIS spatial database schema and setup		✓	
Standardized hazard-record API contract (output format consumed by Decision Support)		✓	
Flood severity & impact prediction model			Sinhalagoda S.A.U.R.A.V
Joint hazard scoring & priority-based alert generation			Prathibhani Y.L.O.L
Resource-aware dispatch & route optimization			Palihena M B

5.USER REQUIREMENTS

5.1 Evidence-Gathering Activities

Stakeholder discussions and domain-expert consultation: A formal data request letter was submitted to the Director General of the Disaster Management Centre (DMC), following guidance from Ms. Samanmali Matharaarachchi (Assistant Director, R&D), covering historical landslide records, rainfall data, and soil information for the landslide, decision-support, and emergency-response components. This engagement surfaced two recurring institutional pain points directly relevant to the Landslide Prediction component: (i) landslide-related data is currently scattered across NBRO, the Department of Meteorology, and DMC with no unified [16] access point, and (ii) existing hazard information is issued at a broad, regional or district scale rather than at the village/DS-division level that would be actionable for evacuation decisions.[15]

Domain-expert/literature evidence: Brenning's comparative review of landslide susceptibility classifiers highlighted that many published models report only "apparent" (training-set) error rates, which are misleadingly optimistic, and recommended models validated on independent, spatially separated test data to avoid overfitting [Brenning, 2005]. This directly informs a requirement for out-of-sample validation in the proposed model. Separately, the Kegalle District AHP-based study demonstrated that combining multiple triggering factors (rainfall, slope, soil, land cover, distance to river) raised classification accuracy from 50–70% (rainfall alone) to 86% (multi-factor) [Gunasinghe et al., 2023], evidencing user/institutional need for multi-factor rather than single-factor prediction.

Design-thinking / accessibility consideration: Reviewing NBRO's existing public tools (e.g., LIRIS) and government early-warning practice showed that current landslide-hazard outputs are typically presented in technical, English-only formats, which are not usable by elderly residents or non-technical village-level authorities in high-risk rural districts such as Nuwara Eliya motivating the localization and accessibility requirements below.

5.2 Translated System Requirements

Functional Requirements

1. The system shall ingest historical landslide records, rainfall data, slope/soil data, and land-use data from NBRO, the Department of Meteorology, and open geospatial sources (SRTM, OSM).
2. The system shall compute a landslide probability score for a given location using a multi-factor model combining rainfall intensity/duration, slope geometry, and soil saturation.

3. The system shall classify predicted landslide risk into discrete categories (e.g., Very Low, Low, Moderate, High, Very High), consistent with established susceptibility-mapping conventions.
4. The system shall generate an explainable breakdown of each prediction, showing the relative contribution of environmental factors (rainfall, slope, soil) versus human factors (road cutting, deforestation, drainage obstruction).
5. The system shall display predictions on a Web-GIS live map at village/DS-division granularity for the pilot area (Ambagamuwa, Kotmale, Walapane).
6. The system shall support map interface localization in Sinhala and Tamil.
7. The system shall output a structured, geotagged hazard record (location, probability, confidence, contributing-factor breakdown) consumable by the Decision Support component via a defined API contract.
8. The system allows authorized users (e.g., NBRO/DMC officers) to view historical prediction accuracy against confirmed landslide events for a given area.

Non-Functional Requirements

1. **Response time:** The system shall return a landslide-probability prediction for a queried location within 5 seconds under normal network conditions.
2. **Reliability:** The prediction model shall achieve a minimum out-of-sample (spatially independent) accuracy of 80%, evaluated using held-out test regions rather than training-area accuracy alone, consistent with the extra-domain validation approach recommended by Brenning (2005).
3. **Usability:** At least 90% of test users unfamiliar with GIS tools shall be able to identify their village's current risk level on the map interface within 2 minutes, without prior training.
4. **Accessibility:** The map interface shall meet WCAG 2.1 AA color-contrast requirements and support high-contrast display modes for users with low vision, including elderly residents.
5. **Scalability:** The system shall support prediction requests across all DS divisions in the pilot district (Nuwara Eliya) without degradation in response time beyond the 5-second threshold specified above.
6. **Security:** All data exchanged with the Decision Support component via the API shall use authenticated, encrypted (HTTPS/TLS) connections.

7. **Interoperability:** The hazard-record output schema remains stable and versioned, such that changes to the Landslide Prediction component's internal model do not break the Decision Support component's ingestion pipeline.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary market (paying customer): Government disaster-management bodies in Sri Lanka the Disaster Management Centre (DMC), National Building Research Organization (NBRO), and District/Divisional Secretariats in landslide-prone districts (Nuwara Eliya, Kegalle, Badulla, Ratnapura).

Persona 1: “The District DMC Officer”: A mid-level government official responsible for issuing early warnings and coordinating evacuation across several DS divisions. Currently relies on manually compiled rainfall bulletins and static NBRO hazard maps; needs real-time, village-level risk visibility and defensible, explainable justification for evacuation orders.

Secondary market (non-paying, indirect beneficiary): Citizens in high-risk rural villages (e.g., Ambagamuwa, Kotmale, Walapane), particularly elderly residents and those without technical literacy, who need to understand their own risk level in plain, localized language.

Tertiary market (future/adjacent): NGOs and international disaster-response agencies (UNDRR, Red Cross) operating in Sri Lanka, who may license aggregated hazard analytics for programmed planning.

6.2 Value Proposition

For government agencies: a single, explainable, village-level landslide risk system that replaces scattered manual processes across NBRO/DMC/Meteorological, reduces the time from rainfall event to actionable warning, and provides an auditable breakdown of *why* a given area is flagged high-risk (supporting accountability when evacuation decisions are challenged).

For citizens: a free, localized, accessible way to check their own area's risk without needing to interpret technical hazard maps.

6.3 Revenue Model

- **B2G licensing/SaaS subscription:** Annual or multi-year license fee paid by DMC/NBRO (or funded via a government IT procurement budget) for access to the platform, dashboards, and historical accuracy reporting.
- **Grant/donor-funded deployment:** Given the good public nature of disaster response, initial deployment could be funded through World Bank, UNDP, or ADB disaster-

resilience grants, or Sri Lanka's own disaster-risk-reduction budget lines, rather than pure commercial sales.

- **Free citizen tier:** The public-facing Web-GIS map and mobile alerts remain free to citizens this isn't a revenue line, it's the public-good delivery mechanism that justifies government funding.
- **Optional data/API licensing:** Aggregated hazard analytics could be licensed to NGOs or research institutions for a fee, as a secondary revenue stream once the core system is established.

6.4 Cost Estimation

- **Development cost:** Team time (final-year project, largely sunk/academic at this stage); post-graduation, would require ML engineer + GIS developer + frontend developer time.
- **Infrastructure cost:** Cloud hosting (API + PostGIS database + model inference), scaling with number of districts covered estimate based on typical small-scale GIS SaaS hosting (e.g., a few hundred USD/month for a pilot district, scaling per district added).
- **Data acquisition cost:** Largely free/negligible, since core data (NBRO, DMC, SRTM, CHIRPS) is either government-provided under MOU or open-access.
- **Maintenance cost:** Ongoing model retraining, localization updates, and support staff time.

6.5 Pricing Strategy

- **Tiered government licensing:** e.g., a per-district or per-DS-division annual license fee, so a smaller district authority pays less than DMC's national-level access.
- **Pilot-then-scale pricing:** Offer the first pilot district (Nuwara Eliya) at no or reduced cost to build a credible case study, then price subsequent district rollouts commercially.
- **Free forever citizen access:** Explicitly non-negotiable citizen-facing information should never be paywalled, both ethically and because government buy-in depends on public benefit being visible.

6.6 Competitive Advantage

- Multi-factor + explainable prediction, versus NBRO's current LIRIS system, which is largely informational/static rather than predictive and explainable at village level.
- Localization and accessibility (Sinhala/Tamil, elderly-friendly UI) a gap not addressed by existing technical hazard-mapping tools.
- Integration into a broader multi-hazard platform (via the Decision Support and Emergency Response components), rather than being a standalone landslide tool this is

your differentiation versus single-purpose academic landslide-susceptibility studies like Brenning (2005) or the Kegalle AHP study (Gunasinghe et al., 2023), which produce static maps but no live decision-support pipeline.

6.7 Intellectual Property Considerations

- The trained ML model, explainability configuration, and Web-GIS interface code constitute the team's own IP, developed independently of the underlying open data sources.
- Underlying datasets (NBRO, Department of Meteorology, SRTM, CHIRPS) remain the property of their respective source organizations/agencies; commercialization would require formalizing data-usage terms with NBRO/DMC beyond the current academic data-request letter likely a data-sharing MOU rather than outright data ownership.
- Any government deployment would likely require a licensing agreement clarifying whether DMC/NBRO receives exclusive rights, a non-exclusive license, or co-ownership, given they are both the data provider and the primary customer this is worth flagging explicitly in the report as an open consideration rather than assuming a clean-cut answer.

7.BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Laptop/workstation (existing, team-owned)	0 (sunk cost)	Model training, GIS digitizing, and frontend development performed on existing personal devices; no new hardware purchase required for the academic phase.
GPU compute (cloud, pay-as-you-go, if needed)	5,000–15,000 (one-time, during model training phase)	XGBoost/LightGBM training on tabular geospatial features is CPU-feasible; GPU only needed if experimenting with deep-learning alternatives (e.g., CNN on raster DEM tiles) budgeted as optional/contingency.
Mobile test device (Samsung/mid-range Android, for Web-GIS UI accessibility testing)	0 (existing personal device used)	Needed to validate the high-contrast, elderly-friendly UI on real mobile hardware rather than desktop browser emulation only.

7.2 Software Licenses

Item	Estimated Cost (LKR)	Justification
QGIS	0 (open source)	Used for GIS digitizing human factors (road cuts, drainage) no license cost as it is free/open-source software.
PostgreSQL + PostGIS	0 (open source)	Spatial database backend for storing hazard records and geospatial layers; no license cost.
Python ML libraries (XGBoost, LightGBM, SHAP, scikit-learn)	0 (open source)	All core modelling and explainability libraries are open source with no licensing fee.
React.js / Tailwind CSS / MapLibre GL JS / Leaflet	0 (open source)	Frontend and mapping libraries are all open sources; no license cost.
i18next / react-i18next and React Native	0 (open source)	Localization framework for Sinhala/Tamil UI support; open source.

GitHub (private repo, student plan)	0 (open source)	Version control and collaboration; covered under GitHub Student Developer Pack.
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7.3 Cloud Services

Item	Estimated Cost (LKR)	Justification
Cloud hosting for API + PostGIS database (e.g., small VM instance)	3,000–6,000/month during active development/testing	Required to host the model inference API and spatial database so the Decision Support component (teammate) can integrate with it during development, rather than relying on local-only testing.
Object storage (satellite imagery, DEM tiles, model artifacts)	1,000–2,000/month	DEM (SRTM), CHIRPS rainfall grids, and trained model files require persistent storage beyond local disk, especially for reproducibility and team access.
Google Earth Engine (data access)	0 (free research/academic tier)	Used for satellite imagery access; free under Earth Engine's non-commercial research license.

7.4 Summary Justification

The budget is deliberately weighed toward human effort rather than hardware or licensing, since the component's architecture relies entirely on open-source tooling (QGIS, PostGIS, Python ML stack, React/Leaflet) consistent with keeping the solution replicable and low-cost for eventual government adoption (Section 6.4, cost estimation). The only recurring real expense is cloud hosting during active development and team integration testing, which is necessary because the Decision Support component depends on querying your API live rather than through static exports.

8. WORK BREAKDOWN STRUCTURE (WBS)

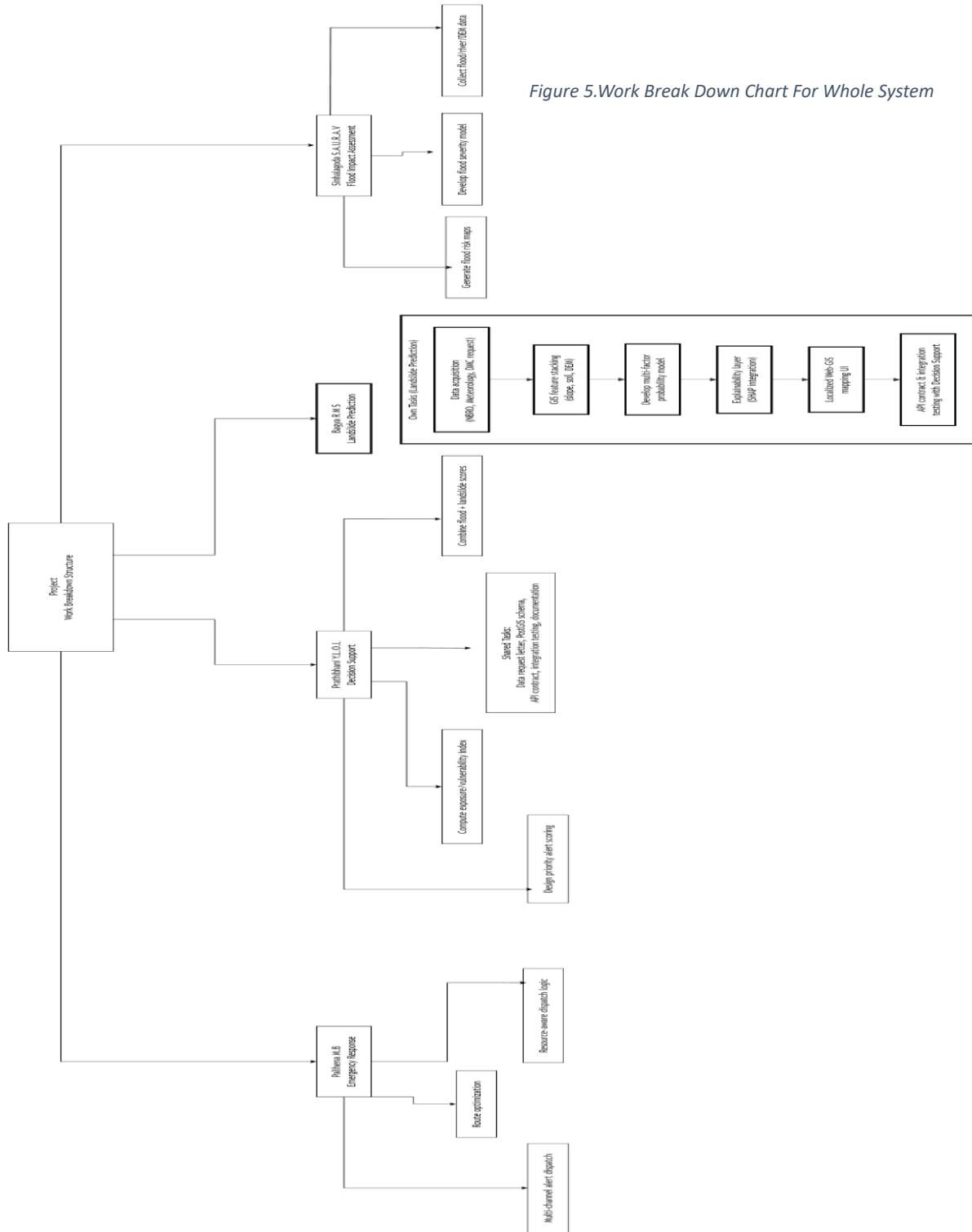


Figure 5. Work Break Down Chart For Whole System

9. GANTT CHART

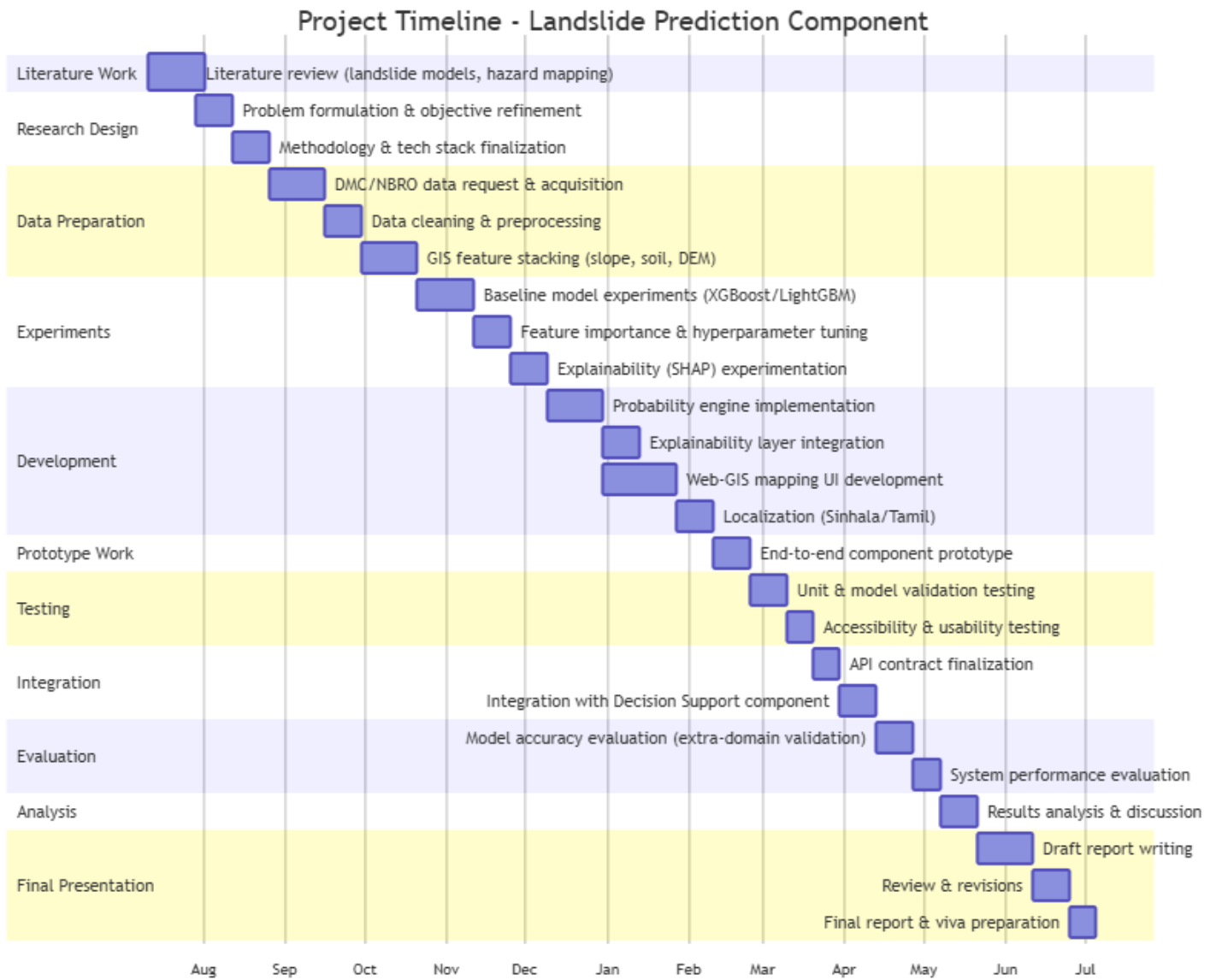


Figure 6. Gaant Chart Of my Component

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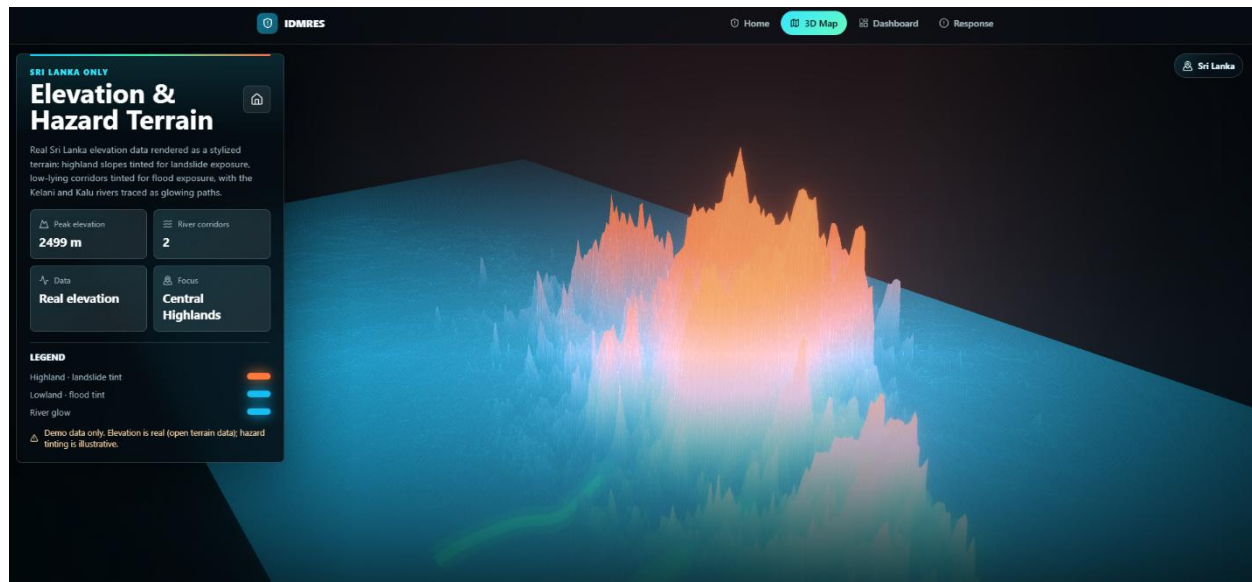
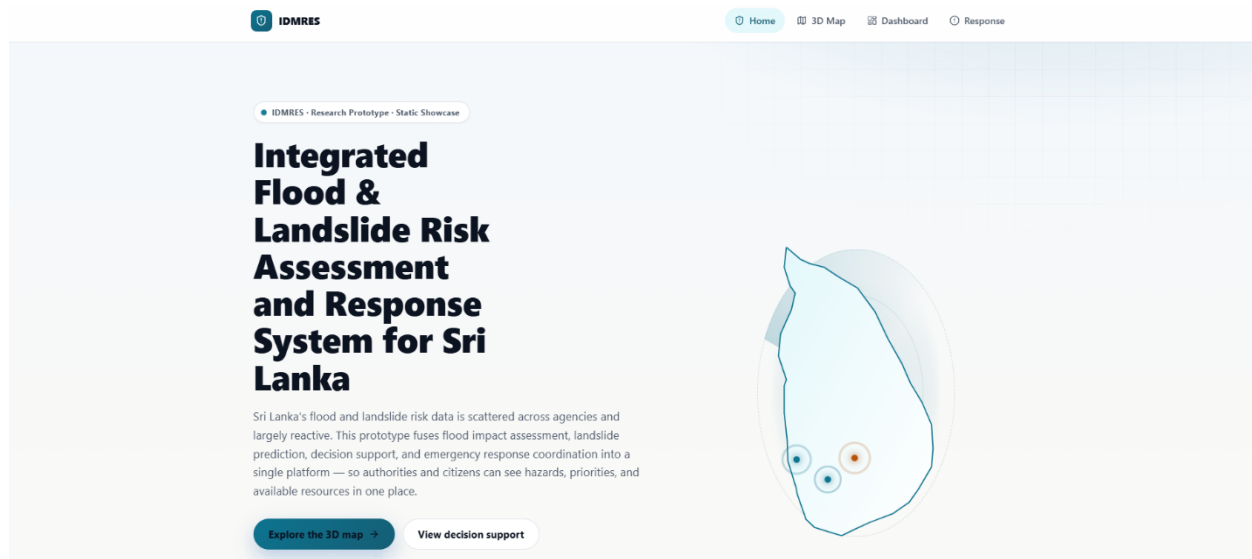
10.APPENDICES

Appendix 1

Appendix A: Prototype Evidence

Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka Live demo:

<https://sri-lanka-3-d-map.vercel.app/>



DECISION SUPPORT

Early Warning & Priority Risk Overview

Joint flood and landslide probabilities, population exposure, and infrastructure accessibility combined into a single priority score per district, translated into graded alerts with recommended actions.

Active alerts

6

Critical risk districts

4

Population exposed

680,000

Avg. model confidence

87%

Joint risk priority by district

Higher score = higher combined flood + landslide priority

Low Medium High Extreme

Colombo					87
Kalutara					87
Ratnapura					87
Nuwara Eliya					87
Gampaha					62
Kegalle					36

Issued alerts

Colombo	Extreme	121 hours ago
Rising flood risk detected across Colombo. Issue evacuation order for exposed low-lying/slope zones		
Kalutara	Extreme	124 hours ago
Rising flood risk detected across Kalutara. Issue evacuation order for exposed low-lying/slope zones		

RESOURCE INVENTORY

Totals across all covered districts, available vs. currently deployed.

Rescue Boats

22 available 18 deployed boats

Ambulances

30 available 24 deployed units

Shelters

16 available 12 deployed sites

Supply Stock

4874 available 2026 deployed kg

Active dispatches

Colombo	In Route	12 min
Rescue Boat + Crew - Rescue Team Alpha		
Kalutara	Deployed	18 min
Ambulance + Medical Team - Rescue Team Bravo		
Ratnapura	Standby	24 min
Rescue Boat + Crew - DDMC Field Unit		
Nuwara Eliya	Returning	On site
Ambulance + Medical Team - Medical Response Unit		
Gampaha	In Route	36 min
Rescue Boat + Crew - Water Rescue Unit		
Kegalle	Deployed	42 min
Ambulance + Medical Team - Rescue Team Alpha		

Evacuation routes

Colombo Evacuation Corridor A	Clear
Route open, moving normally	
Kalutara Evacuation Corridor B	Congested
Heavy traffic reported: expect delays	
Ratnapura Evacuation Corridor C	Blocked
Rerouting via secondary road: avoid this corridor	
Nuwara Eliya Evacuation Corridor A	Clear
Route open, moving normally	
Gampaha Evacuation Corridor B	Congested
Heavy traffic reported: expect delays	
Kegalle Evacuation Corridor C	Blocked
Rerouting via secondary road: avoid this corridor	

Appendix B: Prototype Evidence - Individual Component (Landslide Prediction)

Landslide Risk Framework Component Architecture Prototype

Live demo: <https://sadumina.github.io/Landslide-Suceptability-Component/>

RESEARCH COMPONENT — PROBABILISTIC LANDSLIDE FORECASTING

Reading landslide risk from soil, history and rain

A three-factor machine learning framework for the Nuwara Eliya highlands — built on the premise that where slopes have already failed teaches a model more than the angle at which they currently stand.

Study area: Walapane · Ambagamuwa · Kotmale Core factors: Soil · History · Rainfall Model: XGBoost / LightGBM + SHAP

01 — ARCHITECTURE

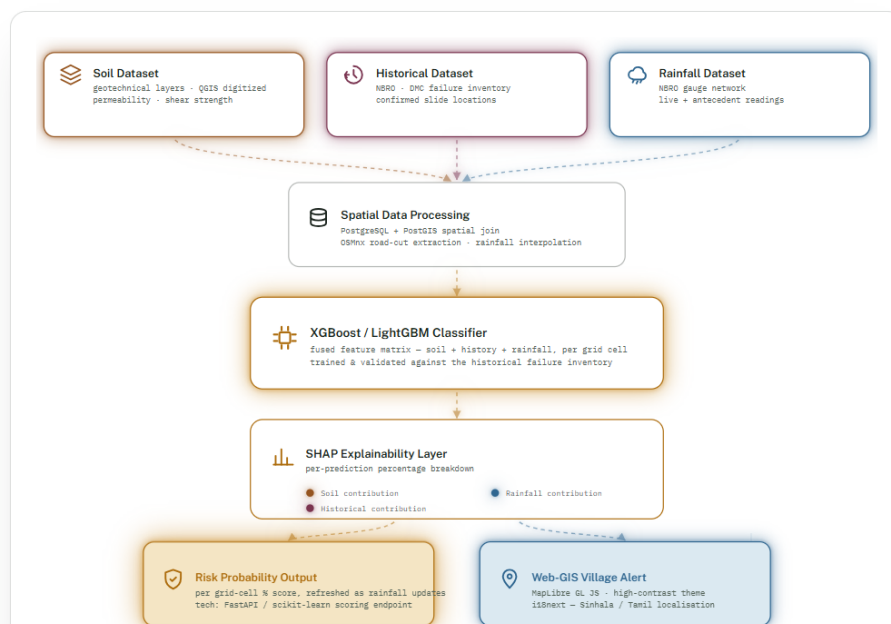
Full system architecture

The complete path from raw datasets to a village-facing alert, in one view: three input datasets, the spatial processing layer that fuses them, the classifier, the explainability layer, and the two things that actually leave the system — a risk score and a map.

01 — ARCHITECTURE

Full system architecture

The complete path from raw datasets to a village-facing alert, in one view: three input datasets, the spatial processing layer that fuses them, the classifier, the explainability layer, and the two things that actually leave the system — a risk score and a map.



Appendix C: Preliminary Research Notes (Requirements Elicitation & Research Design)

[Notes]

Date

$$STI (\text{sediment transport index}) = \left(\frac{1}{22.13} \right) \times \left(\frac{\sin \beta}{0.0396} \right)$$



- + Universal Soil loss Equation (CUE)
- + Sediment Budgets
- + water quality tracking

- ✓ soil type
- ✓ soil thickness
- ✓ rainfall

for according to that research paper best accuracy show RF - 83.7

RF → Random Forest

SVM → Support vector machine learning

for the overall I think RF is best for my forecasting and prediction.

[Notes]

Date

Research paper on STI prediction using machine learning approaches to predict soil susceptibility in Kolathur District, Sri Lanka.

My research component: This one

My component factor: This one has found on whole country for more effective (Colombo, etc.)

Factors

↓
slope, aspect, land use, building, hydrology, topography

Topographic Wetness Index = $\ln \left(\frac{A}{\sin \beta} \right)$ (CIRI)

[Notes]

Date

α (specific catchment area): The total upslope area draining through a specific point per unit contour length.

β (local slope): The local ground slope angle, converted into radians so you can calculate it tangent.

In (Natural algorithm): Used to sort the wide ranging elevation values into a clean, linear number range.

↳ This means the error power of flowing water across a landscape.

Stream power Index = $a \times \tan \beta$

what it is used for and why

- + Soil Erosion mapping → identify or highly vulnerable to topsoil loss
- + Gully detection → It predicts where new drain channels

[Notes]

for Kolathur, Badulla, Kalutara, Kegalle, Nuwara-Eliya, Ratnapura

explainable AI (XAI) → SHAP or LIME

micro level terrain feature

horizontal type → Zone_id, horizontal type, risk score

Phase 01 → Research methodology and Data Acquisition

Phase 02 → The machine learning layers and models

Data sets that I need to get

+ Slope Angle

+ Soil classification → Past - Yellow → non-tarun Req-01

+ Rainfall Intensity + population

+ Soil saturation exposure

+ Dangerous Slope cutting + Infrastructure Accessibility
But Biruq lotus

[Notes]

Date

probability assessment and forecasting of Landslide occurrences



1st question that I have in my mind Is this scope enough for 8-12 months

+ Yes, of course according to researches and notebook in.

Inclusive live mapping → (GIS software)

Complex modeling → Developing probability engine that correlate rainfall with statistics with static variable like Soil Saturation Slope geometry

Appendix 2 - Mandatory Proposal Evidence and Decision Log

Each student should maintain the following log throughout proposal development.

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion

Appendix 3 - AI Use Disclosure

Students should disclose any use of generative AI tools during proposal preparation.

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
Claude AI	Use For Exploring Some Ideas and	Some Get the Idea That To confirm my Thought Using Claude.	Output Is Correct because of Some Parts That I have Taken from Internation Org and National Org and AI gave me the Resources That They Have Didn't Provide external Sites	
ChatGPT	Used For Correct Grammer Mistakes	I put Some Phases that has Lot of grammar Mistakes After I get The corrected Phase Using ChatGPT AI	Output Is Correct	